

Note, This project proposal is given as reference and not as a standard. The document gives the framework. Your project proposal should be precise, instead of copying the text from ChatGpt or other AI sources. The information at point 3 (Benefits etc) should be short in bullet form. Proposal should show that you researched the similar products that led to choice of functionalities for your project (clearly mention the reasoning for the choice of the functionalities). Concept design should be clearly explained with a block diagram. Then explain the choice of hardware, and the reasoning for the choice. Explain the software functions. Identify all hardware required by giving their websites, delivery time (note that delivery time will be more than what stated on the respective websites) so plan accordingly. Before finalizing the hardware, discuss with Andy (andrew.burrell@uwa.edu.au). He may guide you alternate hardware, already available at UWA. Thus saving valuable time.

The project proposal must be submitted in pdf form.
Some bold text below gives the idea of things to be added.

CITS5506: Internet of Things

Project Proposal: Smart Door for Pets

1. Name of project:

Smart Door for Pets

2. Group information:

Group Number: Group 19

N	Names	Student Numbers

3. Why do we want to do this project? What is the problem we will solve?

What benefits will our solution bring and what is its impact?

(1) why this project

Short text in bullets

(2) Problems to solve:

Short text in bullets

(3) Benefits and Impacts:

Short text in bullets

4. Existing Solutions

Automated and smart Pet Doors are currently reshaping the market for pet doors [6]. As technology has developed and production costs decreased, more and more pet owners have expressed their acceptance and expectations for electronic pet doors. Electronic pet doors can be divided into battery-powered and electrically-powered

types, according to their power supply method.

To solve the security problems caused by non-electronic pet doors, existing technologies to identify pets and implement automatic door locks are magnetic tags, microchip sensors, RFID tags, and ultrasonic transmitters.

A report from Virtue Market Research indicates that the battery-powered segment occupied the highest market share in the year 2022 for its convenience that obviates the need for an external power source or any wired connections. Microchip identified is the most popular technologies in the market because it offers amplified security compared to pet doors utilizing collar sensors [7].

We selected the top 5 key players globally in the electronic pet doors market and their own products to make comparisons.

Rank	Product Name / Price	Brand	Company	Material	Target Audience	Identified Technology	Power Supply	Remote Control	Other Features
1	High Tech Automated Pet Door / \$2,000	High Tech Pet	High Tech Pet Products, Inc. (United States)	Glass	small to medium sized pets	RFID collar	a. batter b. electrically	yes	a. Full Manual Control from Front Panel b. Real time notifications c. Operation history d. Custom Timers
2	PetSafe SmartDoor Networked Pet Door / \$540	PetSafe	Radio Systems Corporation (United States)	Plastic	small to medium sized pets	a. Microchip b. RFID collar	a. batter b. electrically	yes	a. multi-point lock b. weatherproof flap seal c. battery life indicator d. current operating mode indicator e. Wi-Fi connection indicator f. schedules to use the door at specified times
3	Microchip Pet Door Connect / \$240	SureFlap	SureFlap Ltd. (United Kingdom)	Glass	small to medium sized pets	a. Microchip b. RFID collar	a. batter b. electrically	yes	a. sends notifications when an intruder has tried to get in b. sends notifications and reports on pet's location and activity c. curfew mode and selective entry for individual pets d. Low battery indicator
4	Electronic Pet Door / \$7,798	PlexiDor	PDUSA, Inc. (United States)	Aluminium	small to medium sized pets	RFID	a. batter b. electrically	unknown	a. The length of time the panel stays open can be adjusted. b. One Touch Programming, easy to add or change codes c. Deadbolt Security, every time the door closes, a steel deadbolt slides into place. d. Certified PDE motors, stainless steel lifting spring and mental gears provide the strength and durability to ensure proper operation in virtually every climate.
5	Wayzn Smart Pet Door / \$499	Wayzn	Wayzn, Inc. (United States)	Glass	small to medium sized pets	RFID	electrically	yes	a. Can work with Alexa b. No harm to the house.

Diagram 1. products comparison

From the comparison, most features of products are similar: being compatible with microchip and RFID, support batteries and adapters' power supply, and support mobile application control remotely.

Some pet door manufacturers such as PetSafe try to differentiate themselves by requiring their products to be environmentally friendly. For example, doors are designed to maintain a balance in temperatures between indoor and outdoor, reducing energy waste, and have wind resistance attribution to prevent outside dust and bugs from entering the room.

Nowadays, artificial intelligence technology has developed, some high-tech companies are beginning to explore a new path to help pets identify such as nose-print recognition in dogs. Petvation proposed a project on Kickstarter [8]. Biometrics technology based on pets' unique information has the advantages of pet friendliness and good operation in identification experience. Through the deep learning analysis model, the key point features of the face and nose-prints are extracted as pet digital identity tags. [9].

When innovation from the product appearance, it is impressive that the Wayzn Smart

Pet Door is actually a smart sensor with a sliding bracket. There is no need to damage the house, but it must be installed on the sliding door. When it is activated, the bracket automatically shrinks to pull back the door body, allowing the pet to pass through [10]. The Pawport smart pet door can slide onto existing non-electronic pet door frames [11].

5. Methodology

5.1 Research and Assessment

- Give outcome (be precise) and short about learning from previous projects and studies with similar functions.
- Tell the reason for the hardware choices.

5.2 Project Scope and Conceptual Design

The product features pet identification, automatic door opening and closing, remote control of the door, object distance detection, pets' behavior reporting and analysis.

In future work, we will add an environmental monitoring system. A temperature and a humidity sensor will be embedded in the pet door, which will automatically be activated at regular intervals or can be manually started through the App, reporting the data to the system. The owner can decide whether the external environment is suitable for the pet to go outside.

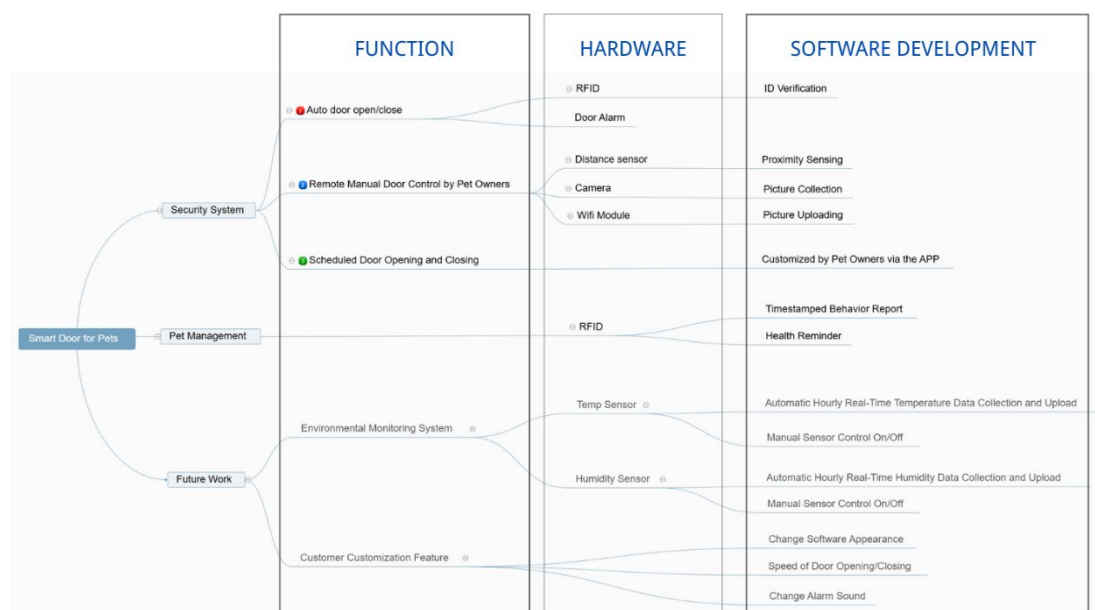


diagram 2. mind map of product features

5.3 Project Timeline

Give project timeline (may be a GNATT CHART)

5.4 Hardwares Selection

- a. Select the Microprocessor: In project scope and design stage, we decided to use Raspberry Pi as the central processing unit. **Explain the reason of the choice.**
- b. Select the RFID Tag: In the project scope and design stage, we decided to use RFID technology. **Explain why you chose?**
- c. Select the Distance Sensor: **What sensor and why chosen? Explain.**
- d. Other Hardwares selection **What hardware and why chosen? Explain. Be precise.**

5.5 Software Development

We will use the flask framework to build the interactive web application, SQLite as a database. This process includes user interface web development, database preparation, back-end logic function development, and API interface development. The web application will include features like pet information storage, door remote control, access recording, data analysis dashboard.

5.6 Testing and Improvement

First, we will test the functionality of each subsystem. After successfully connecting each subsystem, we test the door and the mobile app to ensure that the two can be stably connected, and test whether RFID can be correctly and quickly recognized by the pet door. If the smart door can meet all our expectations in the test, we will further set up extreme environments to test the quality of the smart door, to test whether it can adapt to various scenarios.

6.1 Functionality of Our Solution

Our smart pet door solution prioritizes security problems. For ensure security of houses, there are two important functions in our solution:

- a. Pets Identified: To address wildlife invasion problems, we use RFID technology to allow identified animals to pass in and out, preventing other creatures from illegal invasion. The RFID tags will be designed as a name collar for pets. The RFID reader will be installed inside the pet door. When pets who wear the collar come close, the reader transmits its feedback to the system. If users' information is in the database, the processor will execute the next instruction to open the door. Therefore, only pets with recorded information in the system can enter the house, effectively preventing other creatures from entering.
- b. Automatic Lock: To address illegal encroachment problems, we develop an automatic lock system. When the pet identification succeeds, the system sends the instruction to unlock the door with a prompt tone for pets to pass through. After a certain period (e.g., 2 seconds), the door will lock automatically.
- c. Remote Control: Meanwhile, we plan to develop a web application for pet owners. It allows users to remotely control the state of the smart door, providing flexible door control management, so that anomalous events can be exposed immediately. Users can modify door status and its duration to adapt to different needs and scenarios. For example, the door can be set to open during the day and locked at night.
- d. Objects Distance Detection: To enhance the security function, our solution has an object distance detection function. The pet door is equipped with a camera and a distance sensor. The sensor will be automatically activated every certain period. If the object stays in front of the door for a long time, the camera will automatically open and take a picture, report it as an anomalous event to the central processing system, and send an alert message to owners on the web application. Therefore, the owner can check the picture and message on the mobile phone to find the abnormal situation in time.

Our approach is also helpful in the management issues of pets.

- a. Behavior Report: All pet behaviour data, excluding door control, will be uploaded, saved in a database, and automatically generated into a data analysis report that owners may view on the web application. It aids in creating a more effective schedule for behaviour training dogs.
- b. Health Reminder: The system will notify the owner of a potential health issue with their pet if it compares historical data on the timing and frequency of the pet's entry and exit, for example, if the pet has not used the pet door in a long time based on prior patterns.

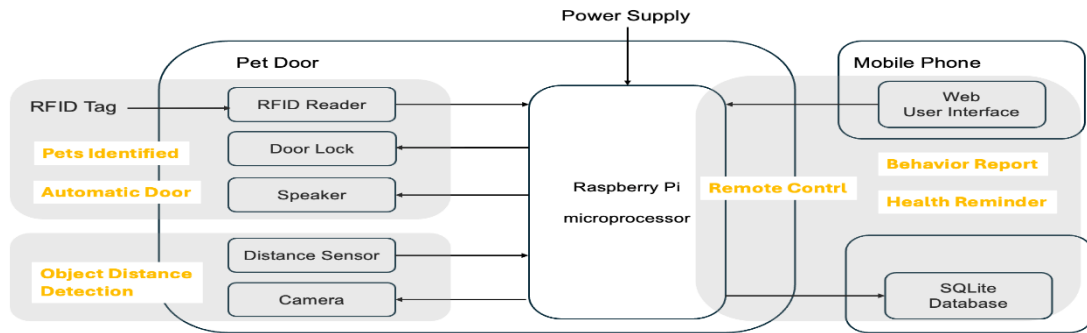


diagram 3. framework of product solution

6.2 Subsystems Process

6.2.1 Pet Identified and Automatic Door Operation

Here, we use **passive RFID tag** as an example.

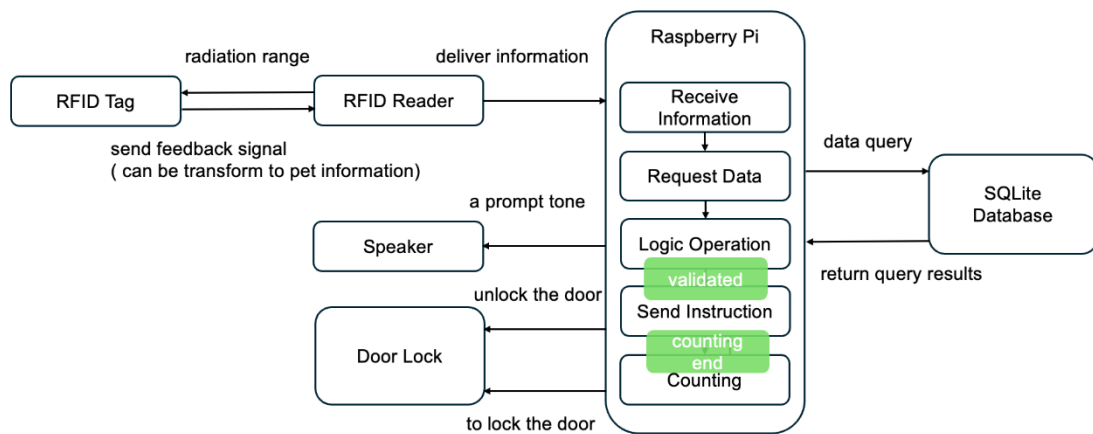


diagram 4. Automatic door lock

- RFID tag receives radio wave when the pet comes in the radiation range of RFID reader. The RFID tag will get energy generated from the RFID reader.
- RFID Tag sends feedback signal to the RFID reader.
- RFID reader receive the signal
- RFID reader transforms the signal to data and submits to microprocessor.
- The microprocessor queries the database to verify pets' information.
- If the information is validated, the door microprocessor sends commands to unlock the electronic lock, allowing the pet to pass through.
- At the same time, the system records the entry timestamp in the database and starts counting.
- After a waiting period, the door microprocessor sends commands to lock the electronic lock.

6.2.2 Object Distance Detection

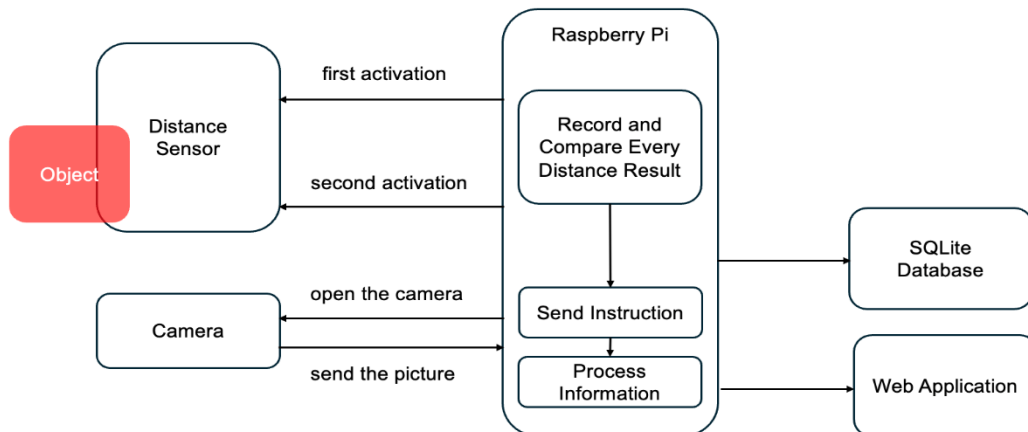


diagram 5. Object Distance Detection

- A distance sensor is set to be activated every certain time and record the first distance.
- The object comes close to the door and stops in front of it for a while.
- Distance sensor is activated and records the second distance.
- The microprocessor compares the number of two distances.
- If they are nearly the same, the microprocessor sends instructions to open the camera.
- The camera takes a picture of the object, then uploads it to the database through the microprocessor.
- The picture will be displayed on the web application.

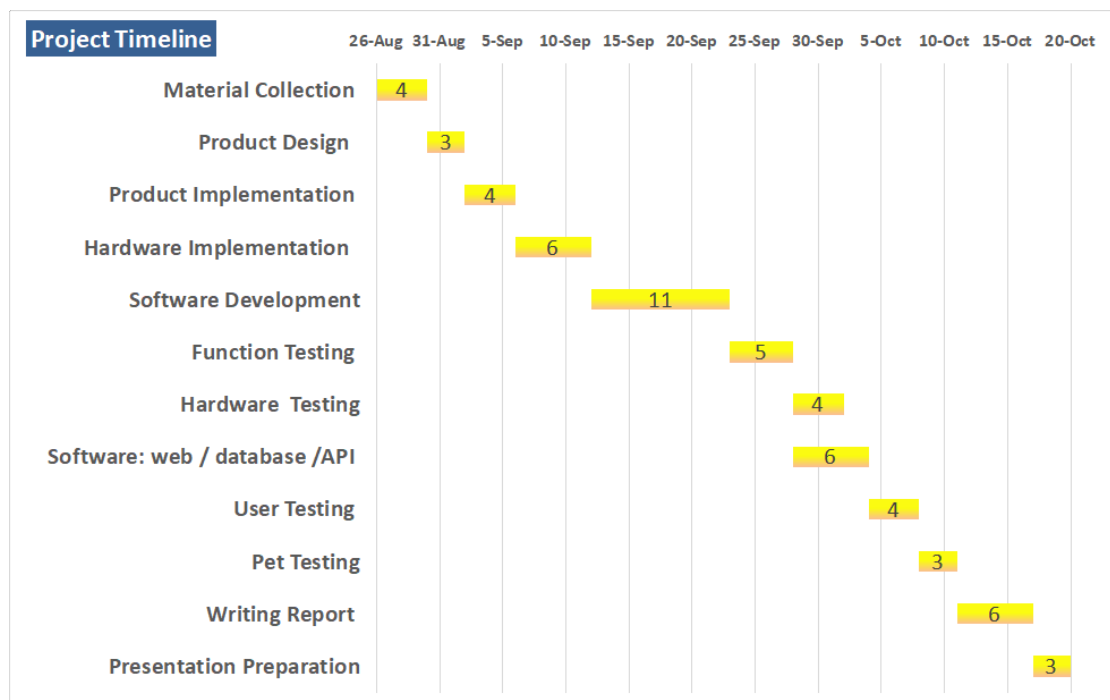
6.2.3 Pet Management

- Behavior Report:** Every time pets' entry and exit will be recorded. The data with timestamp will be uploaded and stored in database. The system will generate the records into intuitive charts for users, such as bar charts, displaying them on the web application.
- Health Reminder:** Weekly pet's entry and exit time and frequency record will be compared to the previous week. If two data have a great difference, the web app will send the user a message automatically with a warning sign and text to remind them to pay more attention to their pet's health recently.

7. Initial Distribution of Work

Name of Student	Student Work Assigned	Reason for the Assignment
Student 1	a. database design and development b. API design development c. business logic implements	a. experienced on software design and back-end development b. deep understanding of the

		system requirements and business process
Student 2	a. RFID identification b. tips sound implements c. automatic door control	a. experienced in hardware design and automating physical components implementations b. understanding of how RFID readers and tags work
Student 3	a. Web application design and development b. remote door control c. Documentation	a. has skills in frontend development b. as a developer involved in the web interface and hardware development, she is well-placed to document these aspects
Student 4	a. distance sensor b. camera activation c. hardware testing	a. experience with hardware sensors b. understanding of sensor triggers and how to activate and manage camera hardware
Student 5	a. RFID identification b. Web application design and development c. Documentation	a. has skills in frontend development b. as a developer involved in the web interface and hardware development, she is well-placed to document these aspects



should be earlier under the heading timeline.

diagram 6. Gantt Chart for the Project Timeline (Days)

9. Hardware Identified

NO.	Items Description	Available at UWA	Cost	Web address	Delivery Time (Days)
1	Raspberry Pi 3 Model A+	Yes	39.95	https://core-electronics.com.au/raspberry-pi-3-model-a-plus.html	4
2	RFID Tag*1	Yes	5.7	https://www.altronics.com.au/p/z6356-mfrc522-contactless-rfid-breakout-module/	2
3	RFID Reader	Yes	12.95	https://www.altronics.com.au/p/s5382-rfid-access-control-key-fob-13.56mhz/	2
4	Rotary Motor	Yes	5.45	https://core-electronics.com.au/feetech-fs90-1-5kgcm-micro-servo-9g.html	4
5	Jumper Wires	Yes	3.20	https://core-electronics.com.au/female-to-female-dupont-line-40-pin-10cm-24awg.html	4
6	Pi Camera	Yes	35.65	https://core-electronics.com.au/raspberry-pi-camera-board-v2-8-megapixels-38552.html	4
7	IR Distance Sensor	Yes	10.25	https://core-electronics.com.au/piicdev-precision-	4

				temperature-sensor-tmp117.html	
8	LED	Yes	3.3	https://core-electronics.com.au/makerverse-led-assortment-5pcs-1.html	4
9	Motor driver	Yes	7.5	https://core-electronics.com.au/makerverse-motor-driver-2-channel.html	4
10	Speaker	Yes	5.19	https://core-electronics.com.au/speaker-3-diameter-4-ohm-3-watt.html	4
11	Power adaptor	Yes	22.80	https://core-electronics.com.au/raspberry-pi-3-power-supply.html	4
12	Bread board	Yes	2.8	https://www.altronics.com.au/p/p1002-400-way-breadboard/	4

10. References

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- [9] Y.-K. Chan and C.-H. Lin, “Dog nose-print recognition based on the shape and spatial features of scales,” Oct. 30, 2023.
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